SCATS · Level 2 · Advanced

Eight mechanisms, one corridor, and the votes that keep them calm

How SCATS turns stop-line counts into cycle lengths, split plans, offsets and linked subsystems, then keeps every green honest second by second.

8 ideas, each building on the last

1 Subsystems and the critical intersection

1 to 10 signals in a typical subsystem

2 Cycle length: bounded steps towards a target DS

+6 / +9 s largest cycle increase per cycle

3 Split plans: equalise DS, then vote

2 of 3 wins a split plan needs before it runs

4 Offsets and green waves

21.6 s offset step for 300 m at 50 km/h

5 Marriage and divorce

18 / 10 link VK to marry / to divorce

6 Tactical control: gap-out, skipping, holding

62.0 s corridor delay with tactical control off (36.9 s on)

7 Public transport priority

29.7 s SCATS bus delay in Lab A1 (fixed 110.8 s)

8 Operating modes and the full system on SUMO

−52% delay cut versus fixed time on the corridor

Three surprises to start

SCATS refuses to obey a single good result.

Even when a new split plan is clearly better this cycle, SCATS waits until it wins 2 of the last 3 cycles. For offsets it waits for 4 of 5. Patience is designed in, because a plan that flips every cycle disturbs drivers more than it helps them.

Junctions get married.

Two neighbouring groups of signals can 'marry' to share one cycle and one clock, and 'divorce' when the traffic between them fades. The thresholds for marrying and divorcing are deliberately different, so they do not argue all day.

Switching off cycle adaptation made our corridor slightly better.

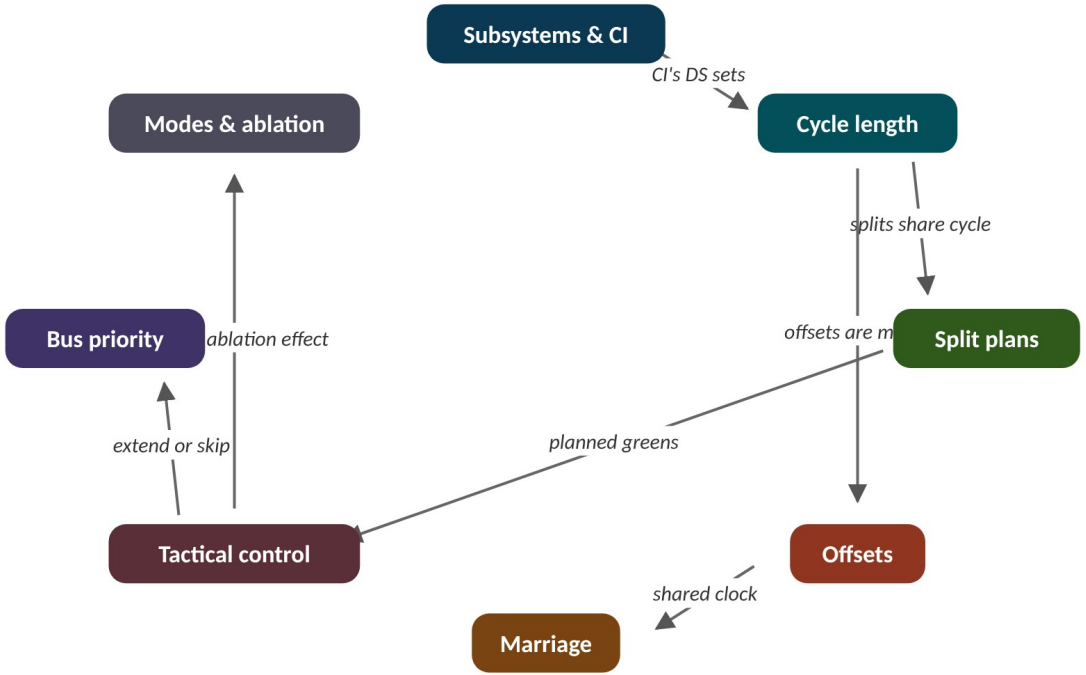
In the SUMO ablation, freezing the cycle at 80 s gave 34.4 s of delay instead of 36.9 s, because 80 s happened to suit that demand. Switching off tactical control, by contrast, raised delay to 62.0 s. Not every mechanism matters equally on every road.

Words you will meet

Subsystem	A group of one to ten signals that share one cycle length so they can be coordinated.
Critical intersection (CI)	The busiest junction in a subsystem. Its DS decides the subsystem's cycle.
DS	Degree of saturation: how much of a green was really used by traffic. 0.9 means 90%.
VK	Reconstituted flow: SCATS's estimate of how many vehicles a green could have carried, from DS.
Split plan	A stored way of sharing the cycle between phases, for example 50% / 20% / 30%.

Offset plan	A stored pattern of start times along a road, for example 'green wave eastbound'.
Strategic control	Decisions made once per cycle by the regional computer: cycle, splits, offsets, linking.
Tactical control	Decisions made every second by the local controller: end a green early, skip a phase.
Hysteresis	Using a higher threshold to switch on than to switch off, so a decision does not flicker.

The map of today's ideas



1 Subsystems and the critical intersection

IN PLAIN WORDS

A rowing crew rows to one rhythm, and the stroke rower at the back sets it. If the stroke tires, the whole boat slows. The other rowers keep the same rhythm but each pulls with their own strength.

- Signals grouped into small subsystems
- Critical intersection: the busiest one
- CI's DS sets the shared cycle
- Others keep own splits and offsets

1 to 10

signals in a typical subsystem

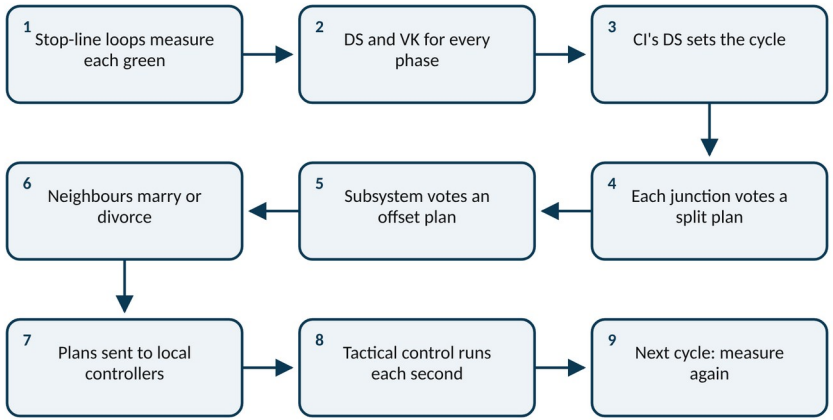
Worked example: Which junction is critical?

$$DS = (g - (T - n \times h)) \div g$$

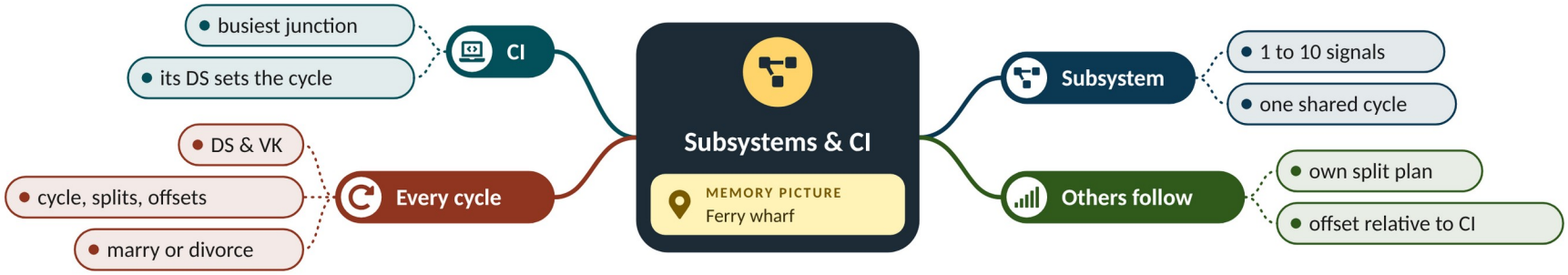
Subsystem J1-J2-J3. At J2 the main phase had 30 s of green, the loop was empty for 22 s in total, 12 vehicles crossed, h = 1.6 s. J1 has DS 0.78 and J3 has DS 0.66.

1. Space a busy stream needs: n × h = 12 × 1.6 = **19.2 s**.
2. Wasted green: T − n × h = 22 − 19.2 = **2.8 s**.
3. Used green: 30 − 2.8 = **27.2 s**.
4. DS at J2 = 27.2 ÷ 30 = **0.91**.
5. Compare: J1 0.78, J2 0.91, J3 0.66. The highest is **J2**.

J2 is the busiest and is the natural CI. Its 0.91 will drive the subsystem's next cycle length.



Remember it: Subsystems & CI



PICTURE THIS · STOP 1: FERRY WHARF

At the wharf, a team of up to ten little ferries is tied together with one rope. The biggest, puffing ferry at the front wears a captain's hat and bangs a drum; all the others row to its beat.

TRAP

The CI is not 'the junction in the middle' or 'the first one'. It is the one whose demand limits the cycle, usually the busiest. Picking a quiet CI makes the whole subsystem's cycle too short.

2 Cycle length: bounded steps towards a target DS

IN PLAIN WORDS

Steering a big ship: you see the harbour entrance, but you turn the wheel a little each minute rather than spinning it. A small, steady correction is safer than one big turn based on one reading.

- Desired cycle = C × DS ÷ 0.9
- Move at most 6 s (9 if overloaded)
- Stay between 40 and 140 s
- Above 110 s, extra goes to main road

+6 / +9 s

largest cycle increase per cycle

Worked example: Two cycles of a rising peak

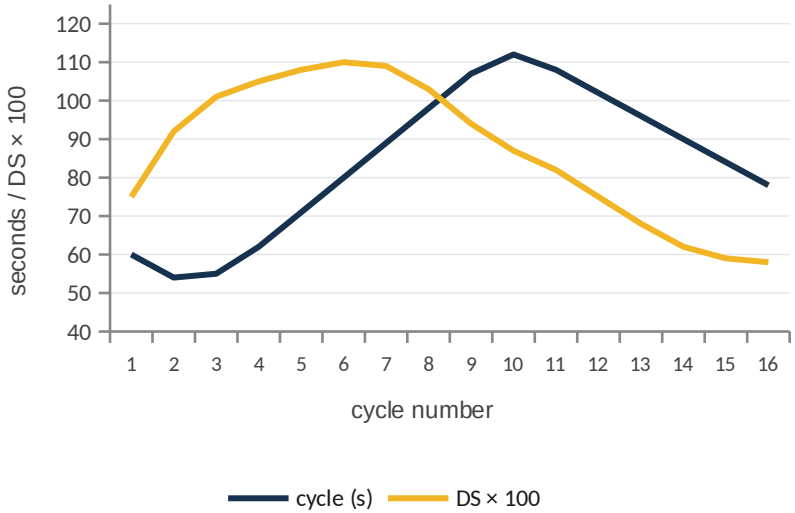
$$\text{desired} = C \times DS \div DS_{target}$$
$$C_{next} = C + \text{clamp}(\text{desired} - C, -6, +6 \text{ or } +9)$$

The CI runs an 80 s cycle and measures DS = 1.05. Target 0.9.

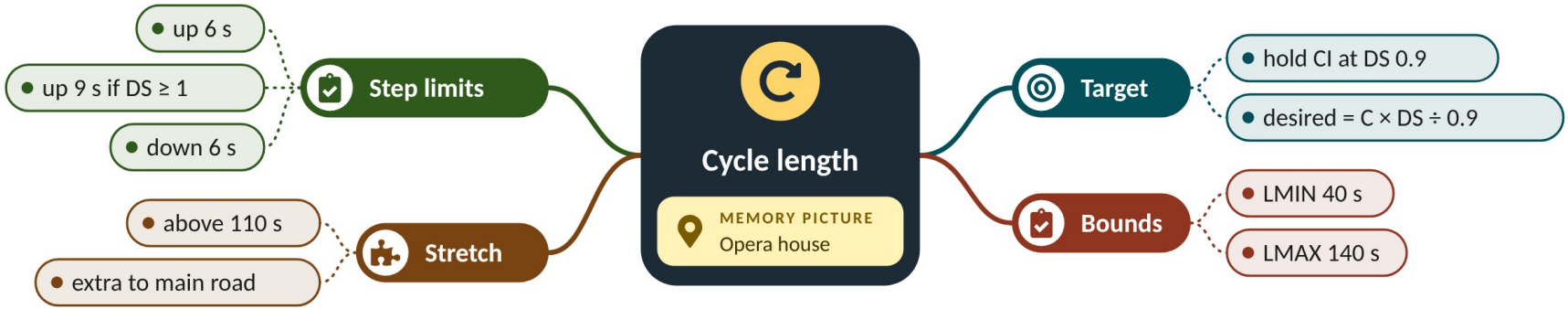
1. Desired = 80 × 1.05 ÷ 0.9 = **93.3 s**.
2. Gap = 93.3 − 80 = 13.3 s. DS ≥ 1, so the step limit is 9 s.
3. Next cycle = 80 + 9 = **89 s**.
4. Next DS = 0.97: desired = 89 × 0.97 ÷ 0.9 = **95.9 s**; gap 6.9 s; limit 6 s (DS < 1).
5. Next cycle = 89 + 6 = **95 s**.

In two cycles SCATS moved from 80 s to 95 s, never by more than its step limit.

A simulated rush hour at the CI.



Remember it: Cycle length



PICTURE THIS · STOP 2: OPERA HOUSE

Inside the opera house a conductor speeds up the orchestra, but a rope on his baton lets him move it only six beats faster at a time, nine if the audience is stampeding. The tempo dial runs from 40 to 140, and a big '0.9' glows above the stage.

TRAP

The step limit is applied to the change, not to the cycle. 'Up 6 s' means Cnext = C + 6 at most, not Cnext = 6. And a low DS does not crash the cycle to LMIN; it walks down 6 s at a time.

3 Split plans: equalise DS, then vote

IN PLAIN WORDS

Sharing pizzas at a party: you look at which table emptied its plate fastest and which left slices, then pick the seating plan that leaves every table equally full. You only move people if the same plan looks best twice out of three times.

- Predict each phase's DS under each plan
- Pick plan with lowest worst-phase DS
- Adopt only after 2 of 3 wins
- Plans are stored, not invented

2 of 3

wins a split plan needs before it runs

Worked example: Choosing among three plans

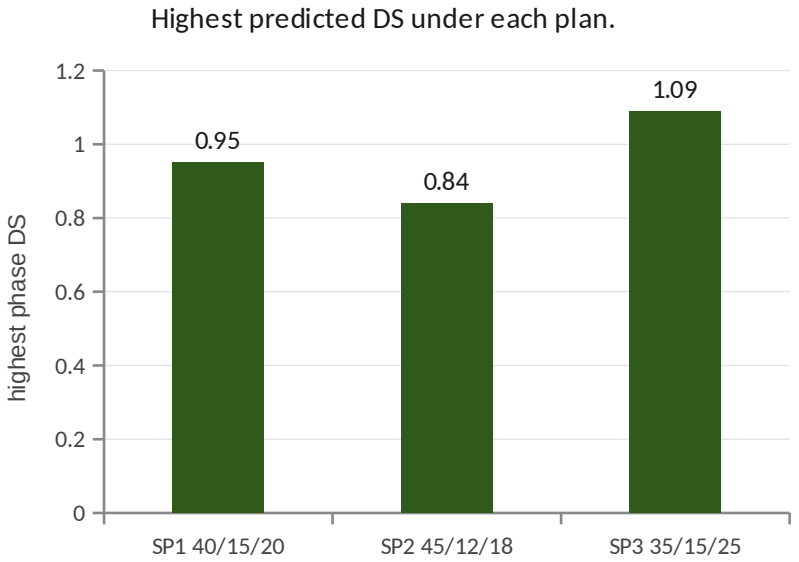
$$DS_{plan} = DS \times g_{now} \div g_{plan}$$

best plan = the one with the smallest $\max(DS_{plan})$

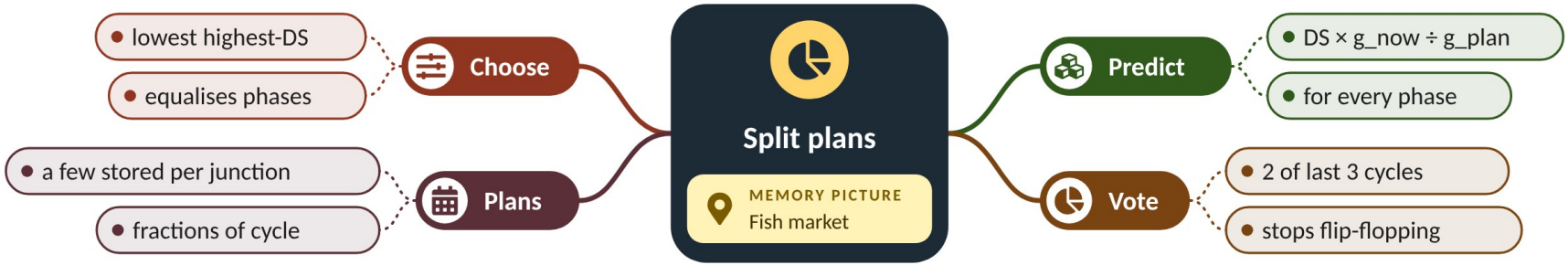
Greens now (plan SP1): A 40 s, B 15 s, C 20 s. Measured DS: A 0.95, B 0.50, C 0.70. SP2 gives 45/12/18 s, SP3 gives 35/15/25 s.

1. SP1 (no change): 0.95, 0.50, 0.70. Highest **0.95**.
2. SP2: A 0.95 × 40 ÷ 45 = 0.84; B 0.50 × 15 ÷ 12 = 0.63; C 0.70 × 20 ÷ 18 = 0.78. Highest **0.84**.
3. SP3: A 0.95 × 40 ÷ 35 = 1.09; B 0.50; C 0.70 × 20 ÷ 25 = 0.56. Highest **1.09**.
4. Smallest 'highest' is SP2 (0.84). SP2 gets this cycle's vote.
5. If SP2 also wins one of the next two cycles, it goes into service.

SP2 takes 5 s from the quiet side road and turn and gives them to the overloaded main road, bringing the worst DS down from 0.95 to 0.84.



Remember it: Split plans



PICTURE THIS · STOP 3: FISH MARKET

At the fish market a stallholder balances three trays of fish on a giant see-saw scale, swapping fish until no tray sags. Before she nails the trays down, three seagulls vote; she needs two squawks for the same tray layout.

TRAP

Do not pick the plan with the lowest average DS. SP3 has a fine average but pushes phase A above 1. SCATS minimises the worst phase.

4 Offsets and green waves

IN PLAIN WORDS

A Mexican wave in a stadium: each section stands up a moment after its neighbour, timed so the wave travels at a steady speed. Time the lights the same way, and a platoon of cars rides the wave of greens.

- Offset = travel time from reference
- Wrap round the cycle (mod C)
- Heavier direction gets the wave
- Adopt after 4 of 5 votes

21.6 s

offset step for 300 m at 50 km/h

Worked example: An eastbound wave on four junctions

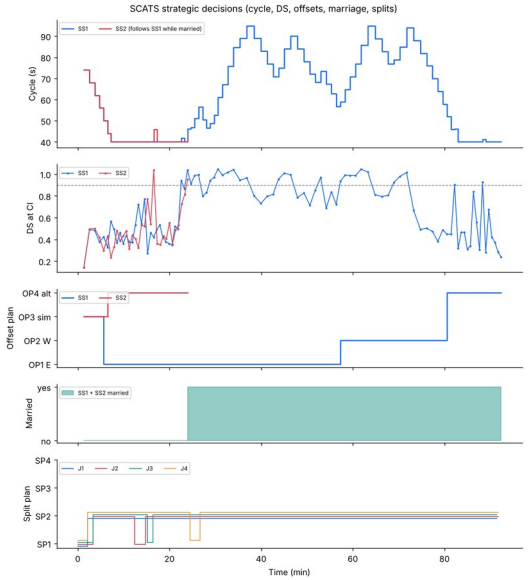
$$\text{OP1: offset}_i = (x_i - x_{ref}) \div v \pmod{C}$$

OP4: offset = C ÷ 2 at every second junction

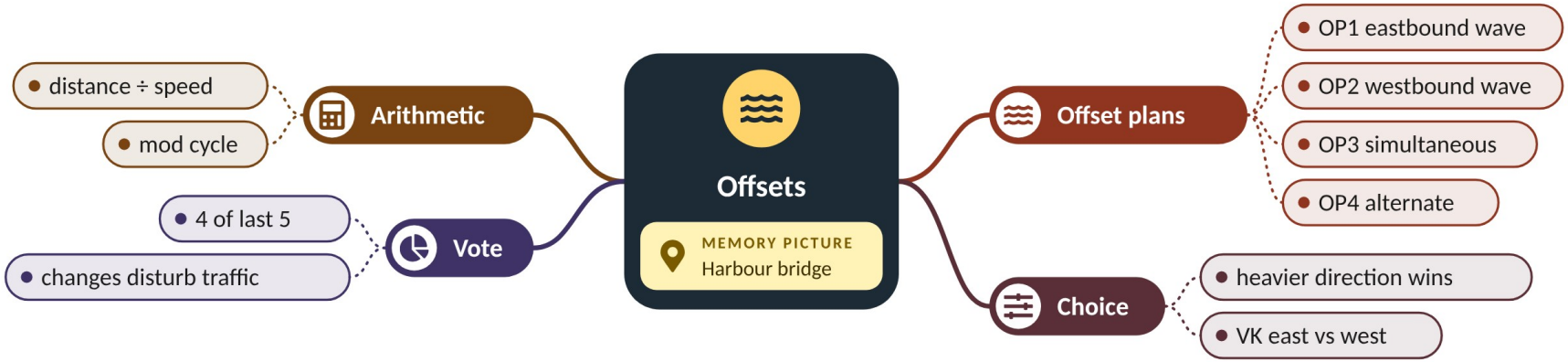
Junctions every 300 m, progression speed 13.9 m/s (50 km/h), cycle 80 s. VK eastbound 20, westbound 12.

1. Travel time per link: 300 ÷ 13.9 = **21.6 s**.
2. OP1 offsets: 0, 21.6, 43.2, 64.7 s (all under 80, no wrapping).
3. OP2 offsets: 0, −21.6, −43.2, −64.7 → mod 80 = 0, **58.4, 36.8, 15.3 s**.
4. Is east clearly heavier? 1.3 × 12 = 15.6, and 20 > 15.6, so **yes**.
5. This cycle votes OP1; it is adopted once it wins 4 of the last 5 votes.

Eastbound drivers at 50 km/h now meet green at every junction; westbound drivers get whatever is left.



Remember it: Offsets



PICTURE THIS · STOP 4: HARBOUR BRIDGE

Along the harbour bridge, giant flags pop up one after another, left to right, exactly as a speeding red car passes each one. A stern referee holds up five cards and needs four to agree before the flags may change direction.

TRAP

Offsets wrap round the cycle. A westbound offset of −21.6 s on an 80 s cycle is the same as +58.4 s. Forgetting 'mod C' gives negative start times that a controller cannot use.

5 Marriage and divorce

IN PLAIN WORDS

A thermostat that turns heating on at 18 °C and off at 21 °C, not at the same temperature. The gap stops the boiler clicking on and off. SCATS uses a similar gap, plus three agreeing readings in a row.

- Married: shared larger cycle and clock
- Marry at VK 18, divorce below 10
- Three votes in a row to change
- Hysteresis stops flip-flopping

18 / 10

link VK to marry / to divorce

Worked example: Tracing link VK cycle by cycle

marry if VK ≥ 18 (and $|C_A - C_B| \le 25$ s) for 3 cycles in a row

divorce if VK < 10 for 3 cycles in a row

Divorced subsystems with cycles 80 s and 90 s. Link VK per cycle: 12, 15, 19, 20, 17, 21, 22, 23, 16, 12, 9, 11, 8, 7, 6.

1. Cycles 3-4 (19, 20) vote 'marry'; cycle 5 (17) breaks the run, count back to 0.

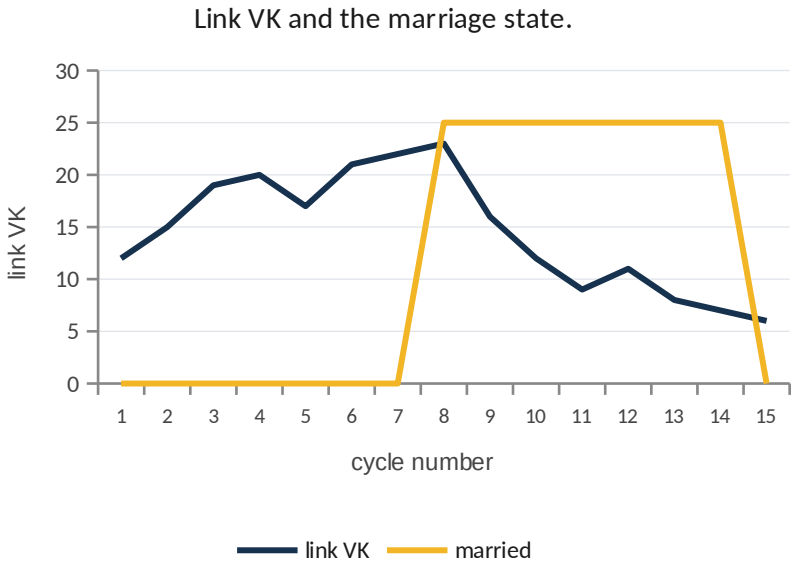
2. Cycles 6, 7, 8 (21, 22, 23): three in a row. **Married at cycle 8**, common cycle 90 s.

3. Cycles 9-10 (16, 12) are above 10: stay married.

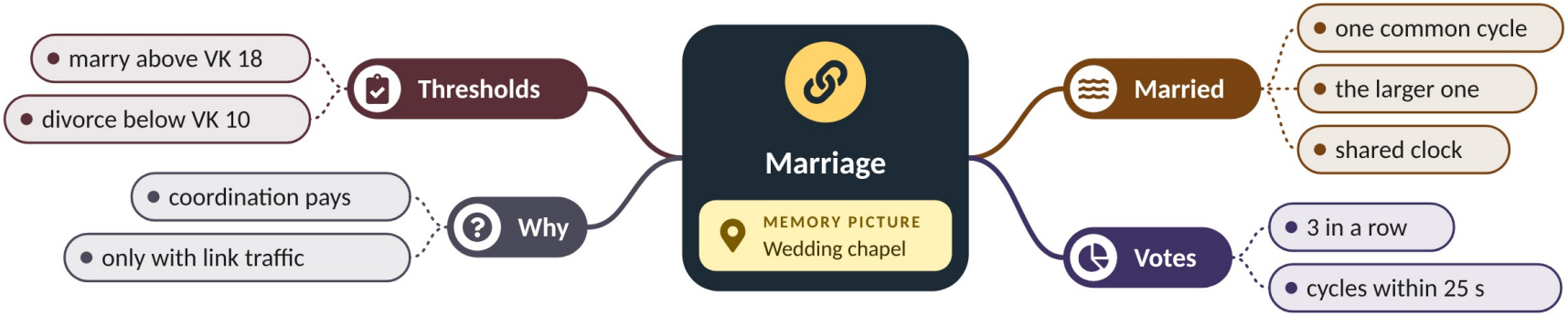
4. Cycle 11 (9) votes 'divorce'; cycle 12 (11) breaks the run.

5. Cycles 13, 14, 15 (8, 7, 6): three in a row. **Divorced at cycle 15**.

Two state changes in fifteen noisy cycles, instead of the many flips a single threshold would give.



Remember it: Marriage



PICTURE THIS · STOP 5: WEDDING CHAPEL

On the rocks above the harbour, two ferries get married in a tiny chapel. The priest needs 18 roses to start the wedding but only takes the rings back if fewer than 10 remain, and three bells must ring in a row each time. The couple share the bigger of their two clocks.

TRAP

Married subsystems do not average their cycles; they both take the larger one, so the busier subsystem is not starved. That is also the cost of marriage: the quieter one runs a longer cycle than it needs.

6 Tactical control: gap-out, skipping, holding

IN PLAIN WORDS

A lift door: it stays open while people keep walking in, and closes once nobody has come through for a few seconds. If nobody pressed the button on a floor, the lift does not stop there at all.

- Gap timer ends a green early
- Never before the minimum green
- Empty phases are skipped
- Main road held to keep the wave

62.0 s

corridor delay with tactical control off (36.9 s on)

Worked example: Does this side-road green gap out?

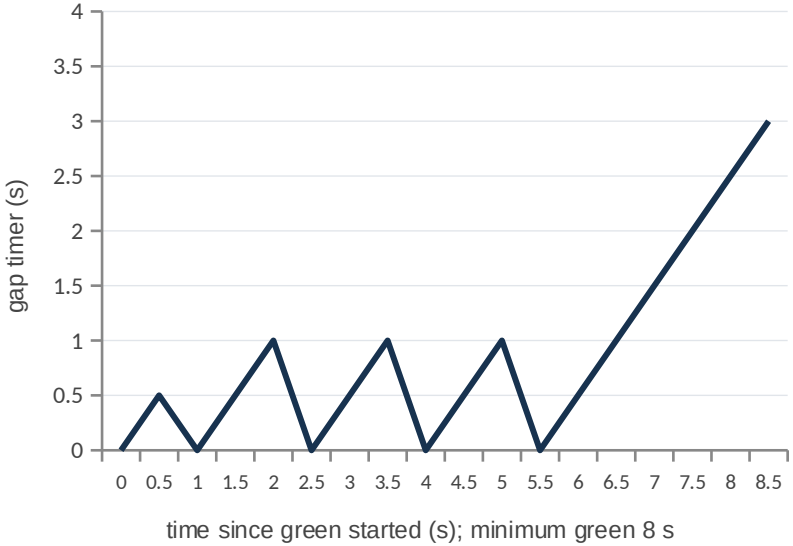
end green when $t \geq g_{min}$ and $(t - \text{last vehicle}) \geq \text{gap setting}$

Planned green 25 s, minimum green 8 s, gap setting 3 s. Vehicles cross the loop at 1.0, 2.5, 4.0 and 5.5 s, then the next one would come at 9.5 s.

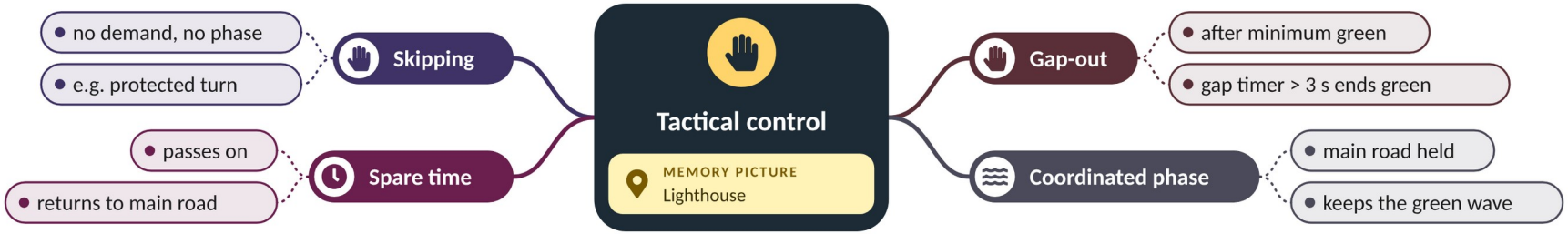
1. Before 8 s the green cannot end (minimum green).
2. At 8.0 s the gap since the last car (5.5 s) is 2.5 s, below 3 s: keep going.
3. At 8.5 s the gap is 3.0 s: **gap-out**. The green ends at 8.5 s.
4. Unused time: 25 − 8.5 = **16.5 s**, passed on to the next phase.
5. The car arriving at 9.5 s waits for the next cycle.

Instead of 16.5 s of empty green, the main road gets that time. This is why switching off tactical control hurt most in the SUMO ablation.

The gap timer during one green.



Remember it: Tactical control



PICTURE THIS · STOP 6: LIGHTHOUSE

In the lighthouse, a lift door stays open while tourists stream in, then slams after three silent seconds, but never before an eight-second bell. The lift skips floors where nobody pressed a button. On the main-road floor a guard props the door open with a giant anchor.

TRAP

Gap-out does not apply to every phase. The coordinated main-road phase is held so the green wave survives, and no phase can gap out before its minimum green.

7 Public transport priority

IN PLAIN WORDS

At a busy shop counter, a customer in a wheelchair (high priority) is served next, someone with one item may be squeezed in early (medium), and everyone else waits their turn (low).

- High: hurry call, serve at once
- Medium: extend or start early
- Low: wait in normal sequence
- Every priority second costs someone

29.7 s

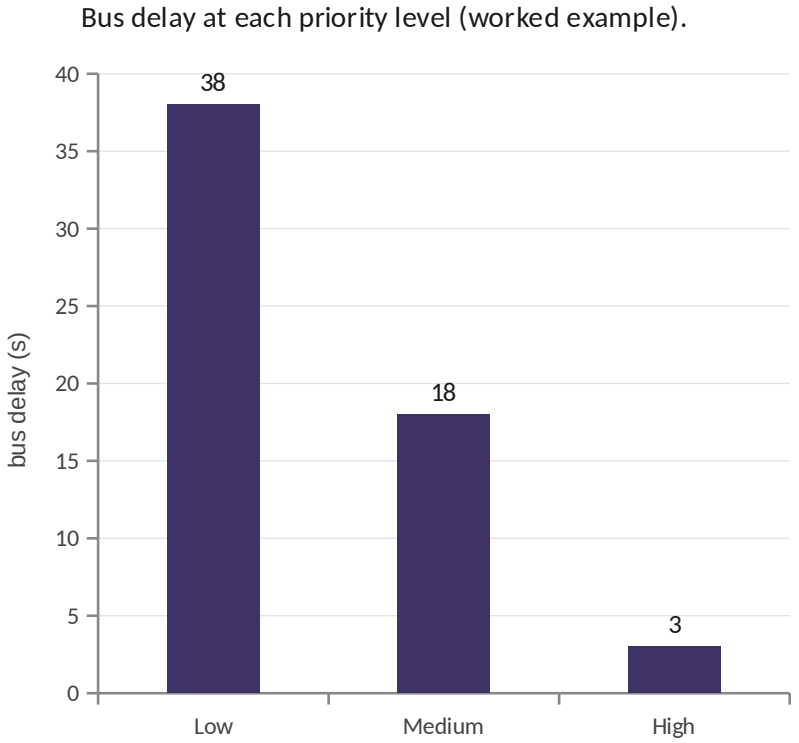
SCATS bus delay in Lab A1 (fixed 110.8 s)

Worked example: A bus that just missed its green

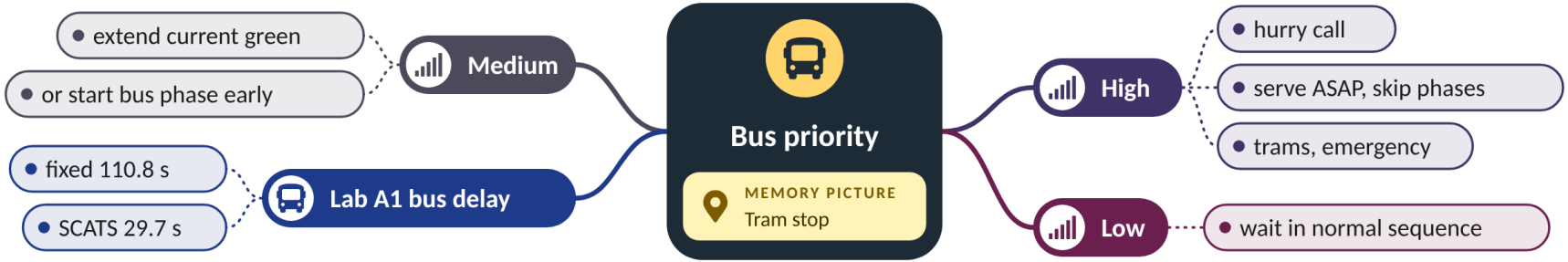
Cycle 80 s: phase A (bus) green 0-40 s, intergreen 5 s, phase B green 45-75 s, intergreen 5 s. The bus arrives at 42 s. Phase B has a 10 s minimum green.

1. **Low:** wait for the next A green at 80 s. Delay = 80 − 42 = **38 s**.
2. **Medium:** cut B to its 10 s minimum: B ends at 55 s, intergreen to 60 s. Delay = 60 − 42 = **18 s**.
3. **High:** skip B entirely; A returns after the intergreen at 45 s. Delay = 45 − 42 = **3 s**.
4. Cost: medium takes 20 s from B; high takes all 30 s.

Higher priority saves the bus more time but costs the side road more. That is why buses usually get medium, not high.



Remember it: Bus priority



PICTURE THIS · STOP 7: TRAM STOP

At the tram stop three vehicles queue for a magic gate. A fire-red tram blasts a hurry horn and the gate flies open at once. A yellow bus waves a ticket and the gate swings early. A grey van just waits its turn, tapping the wheel.

TRAP

Priority is not free. In the ablation, switching bus priority off raised bus delay from 29.7 to 46.4 s but also raised mean delay for everyone from 36.9 to 42.5 s here, because buses ran on the main road with the green wave. On another corridor, aggressive priority can raise general delay instead.

8 Operating modes and the full system on SUMO

IN PLAIN WORDS

An orchestra with a conductor (Masterlink); the same orchestra following printed sheet music and a shared metronome when the conductor leaves (Flexilink); musicians each playing by ear alone (Isolated).

- Masterlink, Flexilink, Isolated, flashing
- Lab A1: 77.2 vs 40.9 vs 36.9 s
- Tactical control matters most here
- Ablation results depend on the corridor

−52%

delay cut versus fixed time on the corridor

Worked example: Reading the ablation

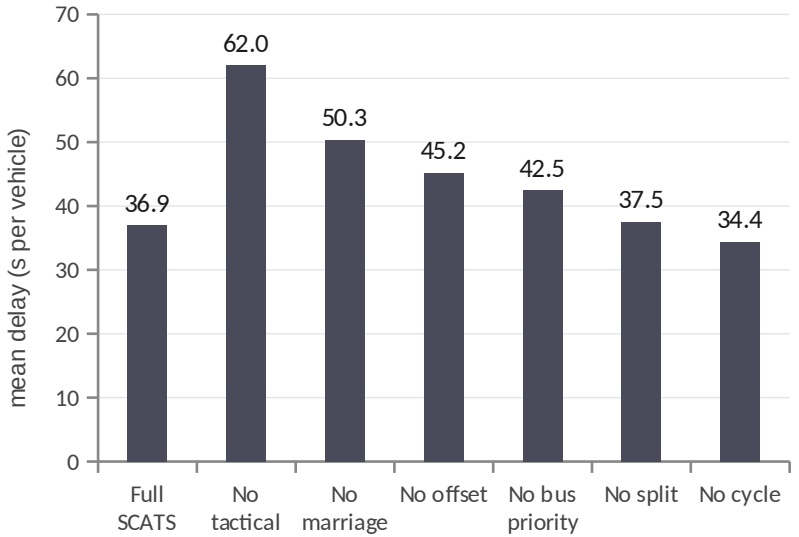
change = (delay without mechanism − delay with all) ÷ delay with all

Full SCATS: 36.94 s mean delay. With one mechanism off: no tactical 62.04, no marriage 50.33, no offset 45.22, no bus priority 42.49, no split 37.50, no cycle 34.42 s.

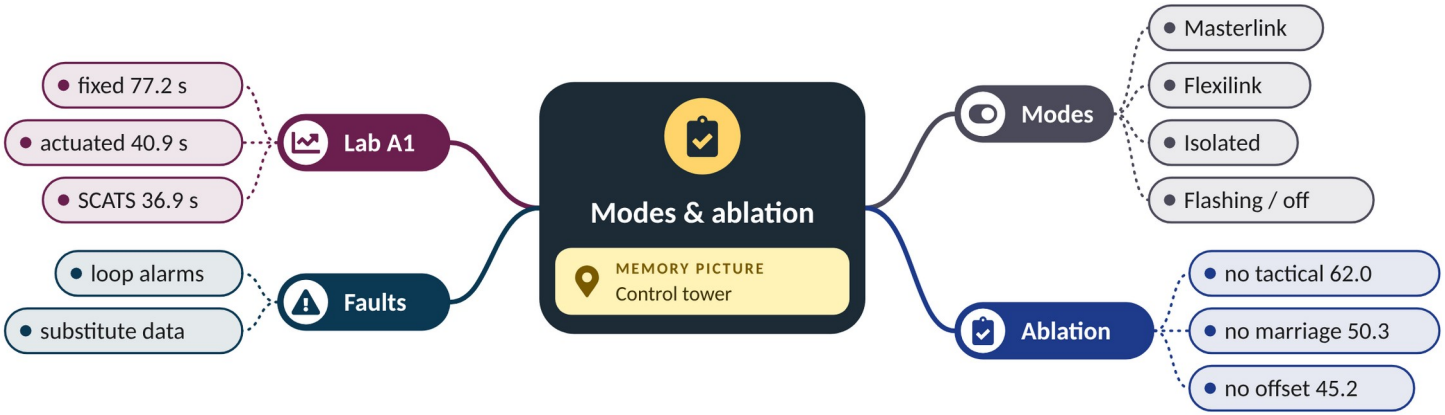
1. No tactical: (62.04 − 36.94) ÷ 36.94 = **+68%**.
2. No marriage: 13.39 ÷ 36.94 = **+36%**. No offset: 8.28 ÷ 36.94 = **+22%**.
3. No bus priority: 5.55 ÷ 36.94 = **+15%**. No split: 0.56 ÷ 36.94 = **+1.5%**.
4. No cycle adaptation: −2.52 ÷ 36.94 = **−6.8%** (slightly better).
5. Versus fixed time (77.2 s): SCATS cuts delay by **52%**; versus actuated (40.9 s) by **10%**.

On this corridor, second-by-second tactical control and coordination (marriage, offsets) carry most of the benefit. An 80 s fixed cycle happened to suit this demand, so cycle adaptation did not pay here.

Lab A2 ablation: mean delay with one mechanism switched off (SUMO, seed 1).



Remember it: Modes & ablation



PICTURE THIS · STOP 8: CONTROL TOWER

In the control tower a conductor leads an orchestra of traffic lights. When she trips over the cable, the lights pick up sheet music and a ticking metronome. Behind her a scoreboard shows seven pillars; the tallest, 62, is labelled 'tactical'.

TRAP

An ablation result is about this corridor and this demand. 'No cycle adaptation was better' does not mean cycle adaptation is useless; it pays off when demand swings far from the fixed value.

Strategic versus tactical control

	Strategic	Tactical
Where it runs	regional computer	local controller at the junction
How often	once per cycle	every second
Decides	cycle, split plan, offset plan, marriage	gap-out, skipping, holding, priority
Damping	step limits, 2-of-3 and 4-of-5 votes, hysteresis	minimum green, gap setting
Ablation (delay when off)	marriage 50.3, offset 45.2, split 37.5, cycle 34.4 s	62.0 s

Things beginners mix up

This…	…is not this	The fact that tells them apart
Split voting (2 of 3)	**Offset voting (4 of 5)**	Offset changes disturb traffic more, so they need stronger agreement.
Married cycle	**Average of the two cycles**	Married subsystems both take the larger cycle.
Lowest worst-phase DS	**Lowest average DS**	SCATS picks the split plan that minimises the highest phase DS.
Gap-out	**Phase skipping**	Gap-out ends a green that started; skipping never starts a phase with no demand.
Flexilink	**Isolated**	Flexilink keeps synchronised clocks, so offsets still hold; Isolated has no coordination.

Check yourself (1 of 2)

C = 100 s and the CI's DS = 1.20 (target 0.9). What is the next cycle?

A. 106 s

B. 109 s

C. 133 s

D. 140 s

Junctions 400 m apart, progression 16 m/s, cycle 90 s. OP1 offset of the third junction (800 m from the reference)?

A. 25 s

B. 40 s

C. 50 s

D. 65 s

Answers are in the speaker notes and in the study guide's answer key.

Check yourself (2 of 2)

Why is the main-road (coordinated) phase normally not allowed to gap out?

A. It has no detector

B. Ending it early would break the green wave

C. It always has demand

D. Rules forbid it

Which pair of mechanisms carried most of SCATS's benefit on the Lab A2 corridor?

A. Split and cycle adaptation

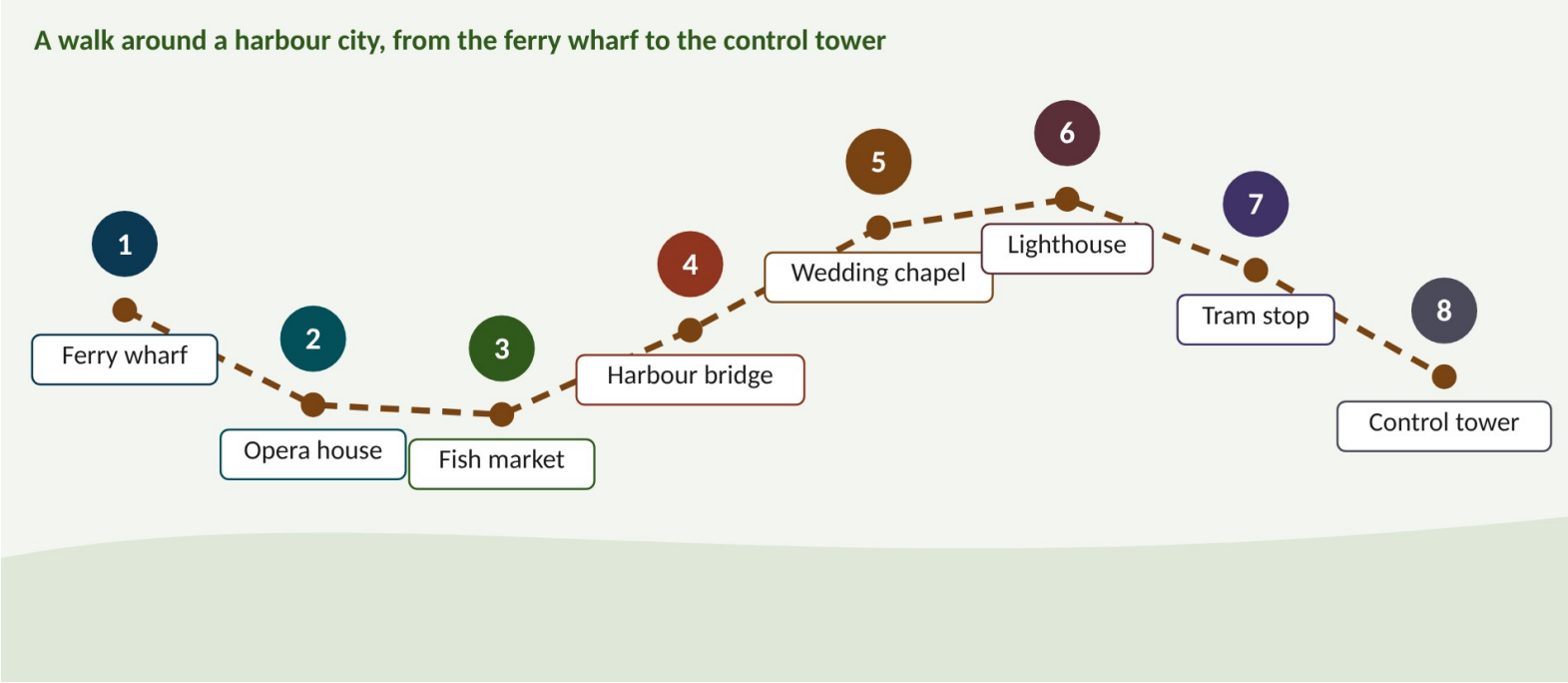
B. Tactical control and marriage

C. Bus priority and cycle adaptation

D. Offset and split adaptation

Answers are in the speaker notes and in the study guide's answer key.

Walk the memory route



1 Ferry wharf = Subsystems & CI · 2 Opera house = Cycle length · 3 Fish market = Split plans · 4 Harbour bridge = Offsets · 5 Wedding chapel = Marriage · 6 Lighthouse = Tactical control · 7 Tram stop = Bus priority · 8 Control tower = Modes & ablation

Numbers to remember

- Target DS 0.9; h = 1.6 s
- Cycle steps +6 (+9 if DS ≥ 1), −6; LMIN 40, LMAX 140, stretch 110 s
- Split vote 2 of 3; offset vote 4 of 5
- Marry VK ≥ 18, divorce < 10, 3 votes, cycles within 25 s
- Gap setting 3 s; main road held
- Lab A1 delay 77.2 / 40.9 / 36.9 s; bus 110.8 / 51.8 / 29.7 s
- Ablation off: tactical 62.0, marriage 50.3, offset 45.2 s

Keep going

Study guide: SCATS_Advanced_Study_Guide.pdf · Interactive labs and SUMO code: smrahmanbd.github.io/scats-learning-lab